Unit I:- Linear Algebra

Topics:

Rank of a Matrix: Definition, examples (page 3.9, J&I), Echelon Form of a Matrix (Page 3.43,J&I)

Problems:

- Exercise 3.1 Q.26 to 30 (page 3.17, J& I)
- Examples 34 Q. 1 to 9 (page 212, CP)
- Exercise 3.3. Q. 1 to 10 (page 3.3, J&I)

Topics:

Consistency of Simultaneous Equations: Solution of system of homogeneous and non-homogeneous system of linear equations using ranks of the matrices, Theorem 3.7.(Page 3.42, J&I)

Problems:

- Exercise 3.3, Q. 26 to 45. (page 3.54, J&I)
- Example 34, Q. 12 to 18 (page 212, CP)

Topics:

Vector Spaces: Definition and examples (page 3.18, J&I), Subspaces, Linear dependence and independence of vectors, Dimension and basis.

Problems:

- Exercise 3.2, Q. 1 to 9. (page 3.37, J&I)
- Exercise 3.2, Q. 21 (page 3.38, J&I)
- Exercise 3.2, Q. 27 (page 3.39, J&I)

Topics:

Eigen Value Problems: Eigen Values and Eigen Vectors (page 3.55, J&I), Properties of Eigen values and Eigen vectors (page 3.57, J&I), Diagonalizable Matrices (page 3.66, J& I)

Problems:

- Exercise 3.4, Q. 7 to 35. (page 3.91, J&I)

Topics:

Cayley Hamilton Theorem: Theorem 3.9. (page 3.57, J&I), verification using examples and its applications.

Problems:

- Exercise 3.4, Q.1 to 6. (page 3.91, J&I)
- Example 35, Q.18 (page 225, CP)

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Course Incharge